

Moving 126,000 devices to Windows 11 across Tesco

One of the largest end-user technology programmes in UK retail history, delivered across stores, distribution centres and corporate offices with Microsoft Azure and Intune.

Scale	Coverage	Platform	Security model
126,000 devices	Stores, distribution centres, offices	Azure AD and Intune	Zero Trust

The challenge

Tesco needed to modernise the computers its people use every day. The estate ran on legacy Windows 10 environments and spanned very different workplaces: busy shop floors, warehouses and head-office desks. Each had its own needs, and all had to meet Tesco's security standards.

The aim was a secure, cloud-managed and scalable desktop environment that supported Tesco's digital-retail strategy. It also had to improve operational efficiency across the whole UK footprint.

Objectives

- Migrate 126,000 devices from Windows 10 to Windows 11
- Bring Azure AD and Intune together for unified endpoint management
- Strengthen security through Zero Trust principles
- Streamline deployment and support across every Tesco site
- Give store and office colleagues a consistent, high-performance experience

The approach

Device management moved under Azure Active Directory and Intune. That gave Tesco centralised control, automated provisioning and real-time compliance monitoring. The work ran in four phases.

Phase 1: Assessment and planning

- Audited the Windows 10 estate for hardware readiness and application compatibility
- Sequenced migration across retail, logistics and corporate environments
- Set up governance and risk frameworks aligned with Tesco's IT security standards
- Designed Azure-based automation for device enrolment and policy assignment

Phase 2: Azure and Intune integration

- Implemented Azure Active Directory for identity management and conditional access
- Configured Intune for provisioning, compliance enforcement and remote support
- Built standard configuration profiles for store, warehouse and office devices
- Enabled self-service deployment and automated patching through cloud policies

Phase 3: Windows 11 rollout

- Ran phased migration waves across all 126,000 devices
- Piloted in high-volume retail sites to validate performance and compatibility
- Coordinated logistics for device refresh, imaging and user transition
- Provided multilingual training and adoption materials for store and office staff

Phase 4: Optimisation and decommissioning

- Retired legacy Windows 10 infrastructure and its management servers
- Verified compliance and performance through Azure dashboards
- Started continuous improvement cycles for endpoint security and user experience
- Handed operations over to Tesco's managed service teams

Piloting in high-volume stores first meant each wave was proven where the pressure was greatest.

Outcomes

- All 126,000 devices migrated across Tesco's UK sites
- Unified endpoint management through Azure AD and Intune
- Stronger security and compliance, aligned with Zero Trust architecture
- Lower operational overhead and more efficient support
- A consistent, modern Windows 11 experience for every employee

Strategic impact

The programme left Tesco with a cloud-connected, future-ready workplace. With identity, devices and policy managed in one place, the business can operate with greater agility, resilience and security across its nationwide footprint.

It also equips colleagues to collaborate seamlessly, respond faster and innovate at scale.

Technologies: Windows 11, Microsoft Azure, Azure Active Directory, Microsoft Intune